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Submission of Article Manuscript,
Electronics Journal

Greetings,

I am writing in support of our manuscript submission to *Electronics Journal*. The title of our manuscript is: *Broadband RF Phased Array Design with MEEP: A 3D-Printed Open-Source RF Horn in the multi-GHz Bandwidth*. This work is the natural extension of our last publication in *Electronics Journal* (J. C. Hanson, *Electronics Journal* **10** 4 p. 415 (2021)). In our previous work, we developed a technique within open-source computational electromagnetism (CEM). Specifically, using the scale-invariance of Maxwell's Equations, we used the photonics tool MEEP to design and simulate broadband RF phased arrays in two and three dimensions.

In this work, we have used open-source tools, including MEEP and open-source CAD programs, to design, fabricate and test broadband RF antennas. Specifically, we have created the first fully 3D printed, broadband, linearly-polarized RF horn in the [5.5-6] GHz bandwidth. Our manuscript includes relevant citations on the topic, design protocols, example code, fabrication technique, and comparisons between measurements and CEM models. Future work includes expanding bandwidth with larger 3D prints, and exploring operational modes of 3D printed RF phased arrays.

We confirm that neither the manuscript, nor any parts of its content are currently under consideration for publication with or published in another journal. All authors have approved the manuscript and agree with its submission to Electronics. We look forward to peer review, and welcome your feedback.

Sincerely,

Jordan C. Hanson